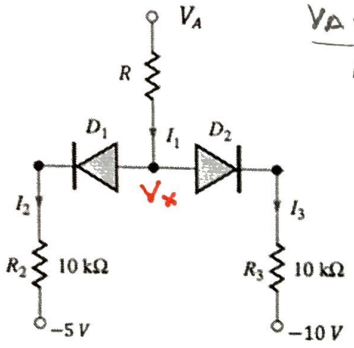
Name: KEY	ID:	$V_A = 10$ V	$R = 3$ kΩ	Mark: /30

1) Find the values of I_1, I_2 and I_3 in the circuit below. Use the ideal diode model. Show all steps. Put answers in table.



Both diodes ON.

$$\frac{V_A - V_x}{R} = \frac{V_x - 10}{10k} + \frac{V_x + 5}{10k}$$

solve for V_x

$$V_x = \frac{V_A - 1.5R}{1 + 0.2R}$$

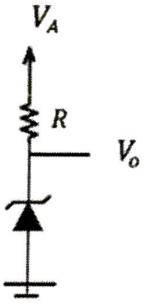
I_1	I_2	I_3
2.19 mA	0.84 mA	1.34 mA

$$I_1 = \frac{V_A - V_x}{R}$$

$$I_2 = \frac{V_x + 5}{10k}$$

$$I_3 = \frac{V_x + 10}{10k}$$

2) The circuit below uses a Zener diode with $V_{ZO} = 3.5\ V$, $r_z = 20\ \Omega$, $I_{z_min} = 1.0\ mA$ and $I_{z_max} = 15.0\ mA$. If the maximum allowed variation in the output is ±0.5%, find the expected variation in the input V_A. No load is connected. Show all steps.



1) Find V_o (without variation)

$$I_Z = (V_A - 3.5)/(0.02 + R)$$

$$V_o = 3.5 + 0.02\, I_Z$$

2) $$\%\Delta V_o = \frac{\Delta V_o}{V_o} \times 100$$

$$\Delta V_o = \frac{V_o}{100}(\%\Delta V_o)$$

3) Find input variation

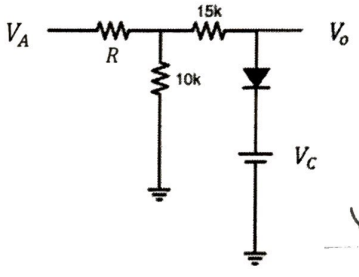
$$\Delta V_o = \Delta V_A \frac{0.02}{0.02 + R}$$

$$\Delta V_A = \Delta V_o (0.02 + R)/0.02$$

4) $$\%\Delta V_A = \frac{\Delta V_A}{V_A} \times 100$$

ΔV_A	$\%\Delta V_A$
± 2.68 V	± 26.8 %

3) For the circuit below, find the output V_o for each value of V_C and record in the table. Take $V_{DO} = 0.7\ V$ for the diode. Show all steps.



V_C	V_o
7.5 V	7.7 V
2.5 V	3.2 V

V_A(V)	V_C(V)	Diode	V_o (V)
5	7.5	OFF	$5(10)/(R+10)$
	2.5	ON	$2.5 + 0.7$
8	7.5	OFF	$8(10)/(R+10)$
	2.5	ON	$2.5 + 0.7$
10	7.5	OFF	10(10)/(R+10)
	2.5 V	ON	$2.5 + 0.7$